A Frame-Only Census of Game Mechanics Built on ARC-AGI-3 Failed Preregistered Transfer Tests on MiniGrid/BabyAI and Procgen

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***Abstract*—I preregistered and tested the claim that a cheap description of game mechanics, read from frames alone and frozen on eight games of ARC-AGI-3, carries over to unseen games from two other benchmarks. It did not. The description, an assumption census, reports for each game whether five mechanical assumptions hold, such as one controlled object. One registered test compared its verdicts with truth tables read from each game's source code; a second asked whether its score predicts on which games three frame-only agents complete the first level. The census and the agents ran on 161 MiniGrid/BabyAI environments under two action sets and on 14 Procgen games, 10,500 runs in all. By the registered outcome table the result is a refutation, decided by the clause that the validity test fails on any benchmark. The census failed that test for three or four of the five assumptions on every tested benchmark and condition, and no agent passed the prediction test. On all six primary MiniGrid/BabyAI lines, a baseline fitted on grid area, action count and color count ranked the games better than the census. An exploratory reading suggests that the probes and the truth definitions were different constructs.**

***Index Terms*—Game-playing agents, general video game playing, benchmark mechanics, preregistration, negative result.**

I. Introduction

This paper reports a preregistered test and its negative answer. Before any confirmatory run I registered the claim that a fixed, frame-only description of a game's mechanics, built and frozen on one benchmark, would describe games from two other benchmarks as their source code does, and would predict, before any agent is run, on which of those games agents that rely on the described mechanics finish the first level. I tested both halves on MiniGrid/BabyAI and Procgen. By the outcome table registered in advance, the claim is refuted: the description failed the validity test on both benchmarks, and its score passed the prediction test for no agent.

The question comes from agent evaluation on unseen games. A competition that scores agents on games they were not built for [1] needs a cheap way to say, before any agent is chosen, what the new games share with the old ones. The agents studied here are frame-only: they see the screen and are told when a level or the game ends, and receive no score, no object list, no text and no game code. They carry built-in assumptions about how games work, for example that one object on the screen is the player and that the arrow keys move it one cell at a time. Knowing which assumptions an unseen game satisfies would predict where such agents finish the first level.

The assumption census was meant to supply that description. It is a fixed set of probes that reads frames only and reports, for each game, whether five mechanical assumptions hold. A1, input coupling, says that the actions change the picture and that different actions change it differently. A2, one controlled object, says that exactly one object on the screen moves as a fixed function of the action. A3, lattice motion, says that each movement action shifts that object by a constant step whenever it is not blocked. A4, blocking by a fixed obstacle set, says that whenever a move has no effect, a fixed set of obstacles explains it, and that this set changes only through the agent's own actions. A5, place goal, says that a level ends when the controlled object enters a particular place, with no other condition at that moment. Each assumption is a property of the game that can be read off its source code. The census tries to infer them from play.

I set the census's parameters while developing a planner on eight of the 25 public games of ARC-AGI-3 [2], and then froze them. The registered protocol states that ARC-AGI-3 "is the development instance and does not count toward generality". Transfer was to be shown on two benchmarks the census had never been tuned on: MiniGrid/BabyAI, which joins MiniGrid [3] and BabyAI [4], and Procgen [5]. I preregistered the claim because almost any reading of such an instrument's output on new games can be made to look sensible after the fact.

II. Background and Related Work

A. General Video Game Playing

Genesereth et al. [6] define a general game playing system as one that can accept a formal description of a game and play it effectively without human intervention; their agents receive the rules. The General Video Game AI (GVGAI) framework moved the problem toward games whose rules the agent is not shown. The first General Video Game Playing Competition asked for a single agent that can play many different games, some of them unknown to the entrants when they submitted, and left it to the agents to discover the mechanics of each game [1]. Perez-Liebana et al. [7] describe the framework as games written in the Video Game Description Language and tracks in which agents play unknown games with or without access to a simulation of the game. The agents studied here sit at the end of that range with the least information: no rules, no simulation and no score, only frames and the signal that a level has ended.

B. Evaluating Agents on Unseen Games

An evaluation on unseen games is only as informative as what is known about how those games differ from the ones

an agent was built on. Legg and Hutter [8] take intelligence to measure an agent's ability to achieve goals in a wide range of environments, which makes the range part of what a score means. Chollet [9] argues that skill at a given task falls short of measuring intelligence, because unlimited priors or training data can buy arbitrary levels of skill, and presents the Abstraction and Reasoning Corpus (ARC). Game benchmarks answer the same concern with held-out content: Procgen generates levels procedurally so that agents can be trained and tested on different levels [5], and BabyAI and MiniGrid offer families of grid tasks [4], [3]. None of them describes, game by game, which mechanical assumptions hold; the census was an attempt to do so from frames.

C. Predicting Agent Success From Game Properties

The closest earlier attempt, Bontrager et al. [10], reported a mainly negative result. They read mechanics features from the rule files of general video game playing games. A clustering of the games by those features agreed poorly with a clustering by controller performance, and the authors concluded that the features were not good indicators of performance. A selection agent built on two of the features did beat each of its constituent controllers on the same games. One possible objection to that design is that features read from rules say nothing about whether an agent could perceive them. Another is that the target, the best controller from a portfolio, was not tied to any assumption the features express. This study was designed to meet both objections. The census measures its features from frames, as an agent would; its accuracy against source-derived truth is tested separately, as test V1; and its prediction target, first-level completion by a stated class of agents that rely on the assumptions, is tested as V2, registered before any run. Two other lines of work predict success from different information. Gaina et al. [11] predict wins on held-out games from an agent's own experience during play, not before it. Laidlaw et al. [12] predict the sample complexity of deep reinforcement learning from a property of the full tabular decision process with its reward, which a frame-only agent does not have.

D. ARC-AGI-3 and Frame-Only Agents

ARC-AGI-3 is an interactive benchmark of turn-based environments that come without explicit instructions; each environment is a series of levels, and at each turn the agent sees a frame, a grid of 64 by 64 cells in 16 possible colors, and takes one action [2]. Its public demonstration set has 25 environments, and the report states that this set does not cover all the mechanics found in the private set. Rudakov et al. [13] published a graph-based explorer for these games, which is one of the agents used here. Liew [14] classifies the public games by the simplest strategy without reasoning that reaches a level, a classification that concerns reachability rather than mechanics and makes no registered prediction. Skoutnev et al. [15] identify goals with a coding agent built on a large pretrained model, which writes an executable world model of each game and so brings a learned prior; that places it outside the prior-free class studied here.

E. Finding the Controlled Object

Bellemare et al. [16] brought contingency awareness into reinforcement learning, defining contingent regions as the parts of an observation whose next value depends on the agent's action, and Choi et al. [17] learned controllable regions with a neural network. A1 and A2 take that idea and replace the learned components with fixed counting rules.

F. Holdout Reuse and Preregistration

When the same held-out set is consulted again and again while choices are made, choices can come to fit it [18]; Roelofs et al. [19] measured this effect across many machine learning competitions and found little of it. The study kept a fixed split of the ARC-AGI-3 games, and kept the two confirmatory benchmarks out of development altogether. Bertinetto et al. [20] describe a model of preregistration for machine learning research, in which the experimental protocol is fixed and reviewed before the experiments are run. This protocol follows that model without the review: it was fixed, tagged in the repository and registered on the Open Science Framework before any confirmatory game was played.

III. Method

The file and line of each registered constant, the full amendment trail, the exclusion lists and the hashes of the frozen files are in the supplement. Repository addresses, commit hashes, registration addresses and preprint identifiers are withheld in this version for double-anonymous review.

A. The Instrument

The census reads the run of the planner N (Section III-D) on a game for 20,000 actions per seed: the frames and the signals that a level or the game has ended, and nothing else. From the planner's report at the end of the run it makes five decisions per seed, with rules written into the registered protocol and implemented in the frozen analysis code. A1 holds when some key has a learned shift and the tried keys' shifts are not all equal. A2 holds when some key was accepted as moving one object and the planner located that object. Acceptance needs three agreeing presses with a margin of 2:1 over the next most frequent displacement. A3 holds when A2 holds and the planner's model of the first level did not end in a failure of prediction or localization. A4 holds when A3 holds and the planner never walked through a color it had classified as a wall. A5 holds when a place recorded at a level end later ended a level again. It fails when such a place was reached without a level end and never confirmed, and it is undetermined otherwise. The verdict for a game is the majority over its 10 seeds: it holds if at least half of the seeds say so, else it fails if at least half say so, else it is undetermined.

I froze every parameter of the census before registration and copied each into the protocol from its source file. They include three agreeing presses to accept a key, the 2:1 margin, at most eight presses per key while learning, a planner search limit of 40000 nodes, and a motion detector that needs at least 2 changed cells. Six files hold the census and its two agents.

Their git blob hashes are registered (Supplement, Section S.6), and the runner refused to start if any of them differed.

I set these parameters while developing the planner on eight ARC-AGI-3 training games, and kept the other 17 public games as a held-out set whose source code I never opened. The split file held 18 games in the held-out set until 21 September 2026, when I moved one of them, dc22, to the development set to debug the agent's wall learning on it. The protocol provided for a truth table derived from source for the eight development games, and for V1 on them, labeled development, once that table existed. I never wrote the table, so the census was never scored against source-derived truth on its development instance. The registered ARC-AGI-3 baseline ran my agents R and N on all 25 public games with 10 seeds and 20,000 actions under the registered code, 500 runs in all.

B. The Benchmarks and How Each Is Presented

The frozen code presses six key identifiers only, so every benchmark is presented through a fixed key map of at most six actions. On MiniGrid/BabyAI (minigrid 3.1.0, gymnasium 1.3.0, Python 3.11.9) there are two conditions. Under primitive actions the keys are turn left, turn right, forward, pickup, drop and toggle. Under absolute moves they are north, south, west and east, each a turn to face that direction followed by a step forward, plus pickup and toggle. The action "done" is offered in neither condition. The grid is rendered with 8×8 pixel tiles, each distinct tile pattern receives an integer index, the four orientations of the agent's triangle are merged into one index, and the grid is placed at the top left of a 64×64 canvas, one row down, because the frozen code reads rows 1 to 62 only.

On Procgen (procgen 0.10.7, Python 3.10, easy mode, sequential levels) the keys are up, down, left, right and two special actions. Frames are 64×64 color images, mapped to integer indices by exact color. The frozen code does not see the top and bottom pixel rows, which was recorded and not corrected. The primary condition draws the whole level into the frame (center_agent off), where the agent can be very small, and the census detects motion only when at least 2 cells change. The protocol therefore registered an operating range before any run, the games whose agent spans at least 2 pixels: nine of the 14 confirmatory games, and Procgen results are reported over all 14 and over these nine. The secondary condition, a view centered on the agent, is identical to the primary one on the ten Procgen games that set the camera option themselves at every level start, nine of them confirmatory and exactly the nine in the operating range; a camera check found 401 of 401 frames identical there. The secondary condition is therefore distinct on five confirmatory games only, and it is reported there and never tested.

C. Truth From Source

Each benchmark has a truth table, committed before the protocol was registered, with one line per environment, read by hand from the installed minigrid 3.1.0 source (per environment class, its step function or instruction verifier) and from the Procgen C++ source at tag 0.10.7. Each entry names the file and line that justify it, and the definitions of A1 to A5 used by the truth tables are written out in the protocol. Under primitive actions A3 fails on every MiniGrid/BabyAI environment, because the displacement of forward depends on which way the agent faces. On the 161 confirmatory MiniGrid/BabyAI environments, A1, A2 and A3 under absolute moves hold everywhere, A4 fails on 5 and A5 holds on 36. On the 14 confirmatory Procgen games, A2 holds on 6, A3 on 2, A4 on 13 and A5 on 6. The truth tables are the registered answer key of test V1; the census is the instrument under test.

D. Agents

Three agents ran on every game, seed and condition. R, my explorer, tries actions and records the frames it has seen; it is evidence for V2. N, my planner, learns a step vector for each key, locates the controlled object, plans toward unexplored places and tests goal hypotheses; it uses the five assumptions by construction, so its V2 lines are an instrument check only, and its runs supply the census. I1 is the graph-based explorer of Rudakov et al. [13], run unchanged behind a boundary adapter, and it is evidence for V2. I1's own code maps five actions, so it has no toggle on MiniGrid/BabyAI and no second special action on Procgen.

E. Registration and Amendment Trail

I registered version v1 of the protocol on 27 September 2026 as git tag prereg-v1 in the study's repository, and on the same day on the Open Science Framework (OSF) as an open-ended registration. It fixed the claim, the agents, the census rules and constants, the truth tables, the benchmarks and their splits between pilot and confirmatory games, the seeds and budget, the tests and thresholds, the analysis code and the outcome table. The registered claim, in the OSF text, reads:

> A fixed, cheap, frame-only probe battery (the assumption census: A1 input coupling, A2 one controlled object, A3 lattice motion, A4 blocking by a fixed obstacle set, A5 place goal), with every parameter frozen on eight ARC-AGI-3 training games, (i) reports the ground-truth mechanics of unseen games at a registered accuracy, and (ii) predicts, before any agent is run, on which unseen games assumption-driven frame-only agents complete the first level. Ground truth is derived from each game's source code and committed before any run. A goal-identification cost G is measured and reported alongside, not tested. ARC-AGI-3 is the development instance and does not count toward generality.

The git-tagged protocol words the claim differently: it names G as part of the battery and says "in three suites"; the tests, thresholds and outcome table are the same in both texts. G counted level ends until one goal hypothesis was left. Its elimination over places is the singletons-plus-empty example of Goldman and Kearns [21], in complement form, and its elimination over features is version-space elimination in the sense of Mitchell [22]. Goal recognition design [23], [24] is related but concerns an outside observer of an actor's goal, whereas G concerns a learner's own goal.

The pilot games, six MiniGrid/BabyAI environments and two Procgen games, could be run before the confirmatory set and are excluded from every confirmatory analysis; the pilot could change adapter code, logged in the next version's dated diff, but never a census constant, threshold, truth entry or frozen file. After the pilot I registered version v1.1 on 27 September 2026 as git tag prereg-v1.1, and on the same day on the OSF. Both registrations have been public since 14 October 2026, when I ended their embargoes. I wrote version v1.2 after the outcome and tagged it prereg-v1.2 on 13 October 2026.

The amendment trail has 16 entries, D1 to D16, given in full in Supplement, Section S.3. D1 to D15 form the v1.1 diff, written on 27 September 2026 after the pilot and before any confirmatory game was played; they fix how each benchmark is presented through the six-key limit, the time caps, the runner, the boundary of I1, the checks, the exclusions from V2, the analysis code, the Procgen camera identity and a replay planned for G. D16, the only entry of v1.2, written after the outcome, withdraws G without measuring it. No threshold, parameter, exclusion list, truth entry or analysis script changed after the prereg-v1.1 tag.

F. Tests and Thresholds

Test V1, instrument validity, compares the census's game verdict with the truth table, per benchmark, condition and assumption, over the confirmatory games. It passes when the share of agreeing games is at least 0.90 and Cohen's kappa is at least 0.70. It fails when agreement is below 0.75, and anything between is "neither". Kappa is computed only where both truth values have at least 3 games; elsewhere agreement decides alone. An undetermined census verdict never counts as agreement, and for kappa it is mapped to the value opposite the truth. That treatment of "undetermined" is in the registered text, which reads "An 'undetermined' census never agrees. For [kappa] it is mapped to the value opposite the truth." (The registered text has the Greek letter where the brackets stand.) The frozen analysis code at the tag implements the same rule.

Test V2, predictive validity, works per benchmark, condition and agent. Each game has a census score, the number of the five assumptions whose verdict holds (undetermined does not hold). It also has a conjunctive prediction, that A2, A3, A4 and A5 all hold, and an outcome: the agent completed the first level in at least half of its 10 seeds. The score is judged by the area under the receiver operating characteristic curve (AUROC) for the outcome. V2 passes when four conditions all hold. The AUROC is at least 0.80. The lower end of its 95% bootstrap interval, from 2,000 resamples of games with seed 20260927, is above 0.60. The balanced accuracy of the conjunctive prediction is at least 0.80. The AUROC exceeds that of a nuisance baseline by at least 0.10. V2 fails when the AUROC is at most 0.65, and the test is void when the share of games completed lies outside [0.10, 0.90]. The nuisance baseline, a logistic regression of the outcome on grid area, number of actions and number of palette colors fitted in-sample, asks whether the census adds anything to the size and variety of a game. Evidence comes from R and I1, and a benchmark fails V2 when every non-void evidence agent fails. On MiniGrid/BabyAI the registered text tests V2 under absolute moves and reports it under primitive actions. On ARC-AGI-3, V2 on the 17 held-out games is reported and not tested.

Test V3, parameter transfer, requires every census constant and frozen file to be identical on every benchmark, registered as a check of the adapter and code fields of every row. The analysis code does not compute it (Section III-J); what the record shows is that the runner compared the six frozen files with their registered hashes before each confirmatory block, refused to start on any difference, and logged all six matching for both blocks.

The protocol fixed three outcomes and the venue that each would allow, before any confirmatory run. Support needed V1 to pass on both benchmarks and V2 to pass for R and I1. Partial needed V1 to pass everywhere with a weaker V2. Refute follows when V1 fails on any suite, or V2 fails on MiniGrid/BabyAI under absolute moves, or V2 is void on every suite. Its registered venue is "repository record only, at most an arXiv note". The table is reproduced in Section IV (Table III).

G. Exclusions From V2

The registered rule (D9) excludes an environment from an agent's primary V2 line when, on every seed, success cannot be reached without an action missing from that agent's map. The action may trigger success, or it may be a precondition of every successful episode. One environment has no success at all and is excluded for every agent. For R and N the rule excludes 6 environments under primitive actions and 49 under absolute moves, leaving 155 and 112. In the pilot, I1, which has no toggle, completed no level on a locked-door environment in any of its 20 runs. D9b therefore applied the same rule to I1's own key map, adding the 57 environments where success needs toggle on every seed. I1's primary lines exclude 63 and 78 environments, leaving 98 and 83. Procgen has no exclusion. Lines with the excluded environments included are reported and not tested. The lists, with identifiers, are in Supplement, Section S.4.

H. Seeds, Budget and Caps

Every agent ran with seeds 1000 to 1009 and a budget of 20,000 actions per game, seed and condition. The wall-clock cap per run was three times the longest pilot run: 3300 s on MiniGrid/BabyAI, where the longest pilot run took 1081.6 s, and 7500 s on Procgen, where it took 2481.5 s. The cap for the ARC-AGI-3 baseline was 3600 s. There were 9,660 confirmatory runs on MiniGrid/BabyAI (161 environments, 2 conditions, 3 agents, 10 seeds) and 840 on Procgen (14 games, 2 conditions, 3 agents, 10 seeds), 10,500 in all. Every run reached its budget, with 0 errors and 0 runs cut by the cap. The MiniGrid/BabyAI block ran on one CPU machine; it started on 28 September 2026 and had finished by 13 October 2026.

I. Checks Before and After the Runs

Four infrastructure checks were registered, and none computes a census or a test statistic. A fidelity check played 200

random actions twice per game and condition and found identical frames in 334 of 334 comparisons on MiniGrid/BabyAI and 32 of 32 on Procgen. A pilot check read the 480 pilot runs and found 0 errors and 0 runs short of the budget, and 0 confirmatory configurations projected over their cap. The camera check is described above. A determinism check compared, on the nine Procgen games where the two camera conditions draw the same frames, every secondary run with its primary run: all 270 of 270 pairs were identical in every field except timing and the condition labels. The analysis code and the determinism check passed 20 of 20 synthetic tests at the tag before the analysis read any confirmatory row.

J. Deviations From the Registered Protocol

Eight departures from the registered protocol, or from what it implies, are recorded here. None changes a test, a threshold or the outcome. The first is D16, written on 13 October 2026, after the outcome was known. The registered claim says that the goal-identification cost G is "measured and reported alongside, not tested", and D16 withdrew G without measuring it, because the instrument it accompanies had failed V1.

The second concerns the venue. The registered consequence of a refutation is "repository record only, at most an arXiv note". On 13 October 2026, with no arXiv endorsement available to me, I recorded that the clause would be read as "at most a preprint" and that no journal submission would follow. After the preprint drafts of 14 October 2026 I reversed the second part and chose to seek peer review of the negative result. An earlier version of this article is a preprint on Zenodo, and this version is submitted to IEEE Transactions on Games. This changes no analysis and no number.

The third is a defect in the frozen analysis code, found on 13 October 2026 and recorded, not corrected. The code computes the two ARC-AGI-3 lines of V2 over all 25 public games, the eight development games included, instead of over the 17 held-out games. It also prints a per-agent verdict on them that has no registered meaning. The fourth is an omission rather than a departure. The pilot runs contained census verdicts for eight games that have truth-table entries, and a V1 comparison on them before the v1.1 tag could have shown the failure before the confirmatory runs. It was not registered and I did not make it; I recorded it as a lesson on 13 October 2026. The fifth I found on 14 October 2026, while writing an earlier version of this article. The registered text tests V2 on MiniGrid/BabyAI under absolute moves and only reports it under primitive actions, but the frozen code prints verdicts, and a suite line, for the primitive-action lines as well. Those verdicts enter no outcome clause.

The last three I also found on 14 October 2026, while reviewing that earlier version against the protocol. Sixth, v1 provided for a truth table of the eight ARC-AGI-3 development games and for V1 on them, labeled development, once the table existed. The record listed the table as still outstanding at the v1.1 tag, and I never wrote it. Seventh, v1 listed I1 among the agents to run on ARC-AGI-3, but the registered ARC baseline ran R and N only. Eighth, V3 was registered as a check of every row's adapter and code fields, and the analysis code never computed it. The ARC-AGI-3 lines are reported and not tested, and V3 matters only for the outcome Support, so none of these three changes the outcome.

IV. Results

The results are the registered tests exactly as the frozen analysis code wrote them; where the code prints a verdict that the registered text does not test, the text says so. Tables I and II give the verdict of each line, and the full tables, with every statistic, are in Supplement, Sections S.1 and S.2.

Fig. 1 and Table I give V1. On MiniGrid/BabyAI under absolute moves, A1 and A2 pass and A3, A4 and A5 fail. Under primitive actions only A1 passes; A2, A3, A4 and A5 fail. On Procgen with the whole-level view, over all 14 games, A1 passes and the other four fail. On the nine in-range games, A1 is "neither", with agreement 0.8889, and A2 to A5 fail. The camera-on condition is reported and not tested. Cohen's kappa is negative for A5 under absolute moves (-0.4125), under primitive actions (-0.3374) and on Procgen (-0.7843), and for A4 under absolute moves (-0.0435).

Fig. 2 and Table II give V2. No agent reached PASS on any benchmark or subset. On MiniGrid/BabyAI under absolute moves, the primary line of R is void, because R completed the first level on 101 of 112 games, a share of 0.9018, above the registered 0.90. I1 is "neither" there, with AUROC 0.7187, and so is N, the instrument check. On the primitive-action lines, which the registered text only reports, the frozen code prints FAIL for R and I1, with AUROC 0.5224 and 0.5370. On Procgen over all 14 games I1 fails, with AUROC 0.6458, and R is "neither", with 0.7222. On the nine in-range games every agent is "neither". On Procgen the lower end of the 95% interval does not reach 0.60 for any agent, which the protocol anticipated, and every line also has a balanced accuracy of 0.5000, below the registered 0.80, so no Procgen line misses a pass on the interval alone.

Under absolute moves the MiniGrid/BabyAI benchmark line is "not failed", decided by I1 alone because R is void, and Procgen with the whole-level view is "not failed" over all games and over the in-range games. The primitive-action line reads "suite FAIL", but the registered text only reports that condition, so it enters no outcome clause (Section III-J).

The nuisance baseline ranked the games better than the census on all six primary MiniGrid/BabyAI lines: the three under absolute moves, which the registered text tests, and the three under primitive actions, which it reports (Fig. 2). Over all 14 MiniGrid/BabyAI lines, primary or not, the baseline is higher on 12. The two exceptions are reported lines under absolute moves with all games included, for N (census 0.6352, baseline 0.5752) and for R (census 0.6434, baseline 0.5589). On Procgen the baseline is below the census on every line of the whole-level view.

The two ARC-AGI-3 lines are reported and not tested, as registered; because of the defect recorded among the deviations, they are computed over all 25 public games, and the "FAIL" printed on them has no registered meaning and enters no gate.

Gate G3 applies the registered outcome table (Table III). The outcome is a refutation, by the clause "V1 fails on any

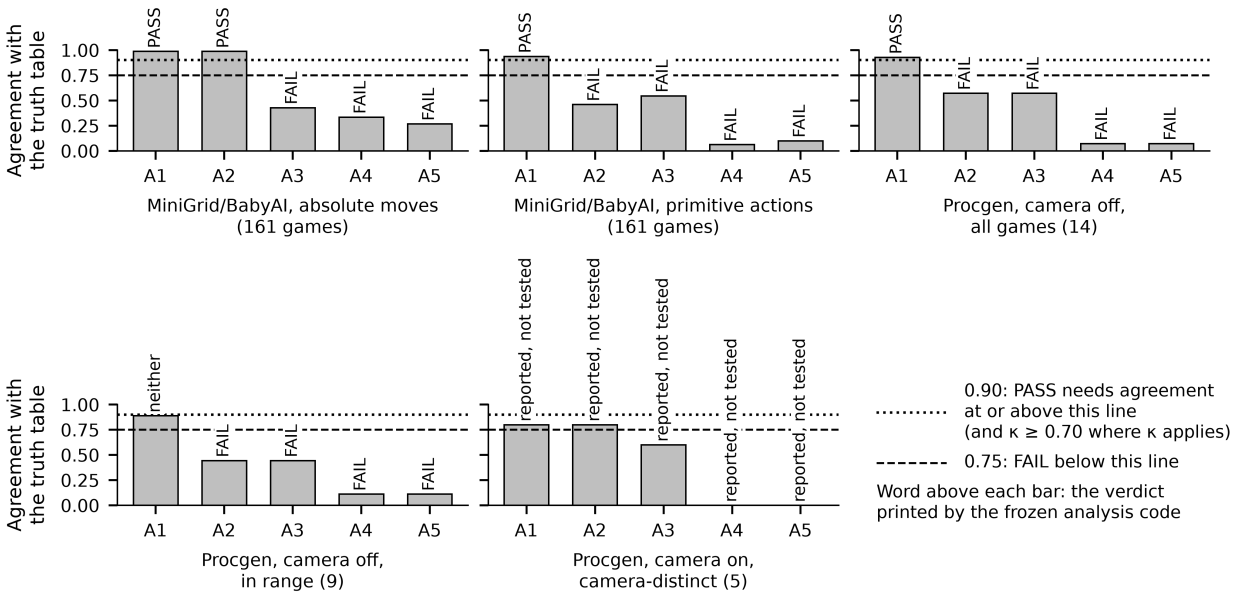


Fig. 1. Test V1: agreement between the census and the source-derived truth table, per assumption, for each benchmark and condition. Dashed line: 0.75, below which V1 fails. Dotted line: 0.90, which V1 needs to pass (with kappa of at least 0.70 where kappa applies). The word above each bar is the verdict printed by the frozen analysis code; a bar of agreement 0.0000 is not visible. "Camera off" is the whole-level view and "camera on" the view centered on the agent. Registered test; the values are those of Supplement, Section S.1.

TABLE I

TEST V1, THE VERDICT OF EACH LINE AS PRINTED BY THE FROZEN ANALYSIS CODE AT PREREG-V1.1, WITH ITS AGREEMENT (THE SHARE OF GAMES WHERE THE CENSUS VERDICT EQUALS THE TRUTH) AND, WHERE BOTH TRUTH VALUES HAVE AT LEAST 3 GAMES, COHEN'S KAPPA (κ). A LINE FAILS BELOW AGREEMENT 0.75 AND PASSES AT 0.90 OR ABOVE WITH κ, WHERE IT APPLIES, AT LEAST 0.70. VALUES UNCHANGED FROM SUPPLEMENT, SECTION S.1.

benchmark and condition	games	A1	A2	A3	A4	A5
MiniGrid/BabyAI, absolute moves	161	PASS 0.9876	PASS 0.9876	FAIL 0.4286	FAIL 0.3354, κ -0.0435	FAIL 0.2671, κ -0.4125
MiniGrid/BabyAI, primitive actions	161	PASS 0.9379	FAIL 0.4596	FAIL 0.5466	FAIL 0.0621, κ 0.0021	FAIL 0.0994, κ -0.3374
Procgen, whole-level view, all games	14	PASS 0.9286	FAIL 0.5714, κ 0.2222	FAIL 0.5714	FAIL 0.0714	FAIL 0.0714, κ -0.7843
Procgen, whole-level view, in-range games	9	neither 0.8889	FAIL 0.4444, κ 0.0000	FAIL 0.4444	FAIL 0.1111	FAIL 0.1111
Procgen, view centered on the agent, camera-distinct games (reported, not tested)	5	0.8000	0.8000	0.6000	0.0000	0.0000

TABLE II

TEST V2, THE VERDICT OF EACH AGENT AND THE BENCHMARK LINE, AS PRINTED BY THE FROZEN ANALYSIS CODE AT PREREG-V1.1, FOR THE TESTED LINES, THE REPORTED PRIMITIVE-ACTION LINE AND THE REPORTED ARC-AGI-3 LINE. THE LINES WITH EXCLUDED ENVIRONMENTS PUT BACK, THE CAMERA-ON LINES AND EVERY STATISTIC ARE IN SUPPLEMENT, SECTION S.2.

line	R	I1	N (instrument check)	benchmark line
MiniGrid/BabyAI, absolute moves, primary	VOID	neither	neither	not failed (decided by I1 alone)
MiniGrid/BabyAI, primitive actions, primary (reported)	FAIL	FAIL	FAIL	suite FAIL (decided by R and I1)
Procgen, whole-level view, all games	neither	FAIL	neither	not failed (decided by R and I1)
Procgen, whole-level view, in-range games	neither	neither	neither	not failed (decided by R and I1)
ARC-AGI-3, 25 public games (reported)	FAIL	not run	FAIL	reported, not tested

suite": V1 fails on MiniGrid/BabyAI in both conditions and on Procgen. The other two clauses of the refutation do not apply, since the V2 benchmark line under absolute moves is "not failed" and V2 is not void on every benchmark. One clause is enough. This classification was recorded on 13 October 2026, before any text about the results was written.

V. EXPLORATORY DIAGNOSIS

This section is exploratory. It asks why an instrument with fixed parameters did not keep its meaning across benchmarks;

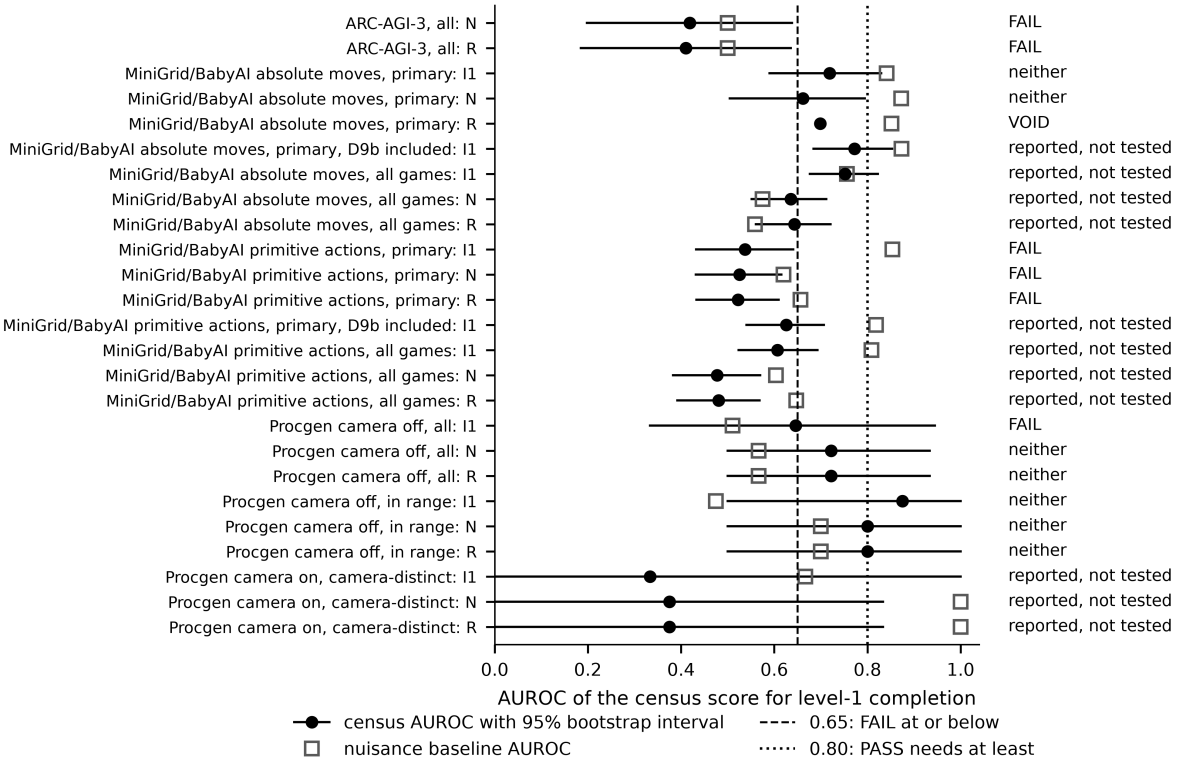


Fig. 2. Test V2: AUROC of the census score for first-level completion (filled circle) with its 95% bootstrap interval, and the AUROC of the nuisance baseline (open square), per agent and line. Dashed line: 0.65, at or below which V2 fails. Dotted line: 0.80, which V2 needs to pass, among three further conditions. The word at the right is the verdict printed by the frozen analysis code; the void line has no interval. The two ARC-AGI-3 lines are reported and not tested, and the "FAIL" printed on them has no registered meaning (Section III-J). The primitive-action lines are reported by the registered text, so the "FAIL" printed on their primary lines enters no outcome clause. "Camera off" is the whole-level view and "camera on" the view centered on the agent; "level-1 completion" is first-level completion. Registered test; the values are those of Supplement, Section S.2.

TABLE III
THE REGISTERED OUTCOME TABLE, AS FIXED IN THE PROTOCOL BEFORE ANY CONFIRMATORY RUN (B: MINIGRID/BABYAI; C: PROCGEN; B-MACRO: MINIGRID/BABYAI UNDER ABSOLUTE MOVES).

outcome	**definition (registered)**	**venue (registered)**
Support	V1 passes on B (both conditions) and C, V2 passes for R and I1 on B-macro and C (not void), and V3 holds	JAIR full article
Partial	V1 passes everywhere, but V2 passes on B-macro only or is void on C, or passes on C for R only	IEEE Transactions on Games full paper
Refute	V1 fails on any suite, or V2 fails on B-macro, or V2 is void on every suite	repository record only, at most an arXiv note

it was written after the outcome was known, tests nothing, and changes nothing above. Its numbers come from an exploratory script that calls the frozen analysis functions unchanged (Supplement, Section S.5); the census verdict stored in each of the 3,500 planner rows equals the verdict recomputed from that row's report, with 0 mismatches. Fig. 3 shows, for every V1 line, how the games split between truth and census, and Fig. 4 how often single seeds disagree with their game's majority. Both are exploratory.

In this exploratory reading, for A1 the census agrees with the truth on 151 of 161 games under primitive actions, 159 of 161 under absolute moves and 13 of 14 on Procgen; every disagreement is a game where the truth holds and the census fails. For A2 under primitive actions, 87 games have the truth holding and the census failing. On their 870 seeds the two turn keys were recorded as producing no movement on all 870: a turn changes the agent's facing but not its cell, and the adapter merges the four facings into one tile index, so a turn changes nothing the census can see. Under absolute moves A2 agrees on 159 of 161 games. On 6 Procgen games the truth fails A2, because other entities also respond to the action, and the census says it holds; the census rule has no clause for that.

For A3 under absolute moves, 92 games have the truth holding and the census failing. Four step vectors were accepted on 850 of their 920 seeds, so the learned motion was mostly right; of the 775 seeds whose verdict is "fails", 755 carry the planner's first-level outcome "mispredicted". The census's A3 failure therefore comes from the planner's prediction during play, not from the learned step vectors. Under primitive actions the truth fails everywhere, and the census says "holds" on 73 games. The census rule for A4 is conditional on A3, so most A4 failures on MiniGrid/BabyAI follow from those of A3; on

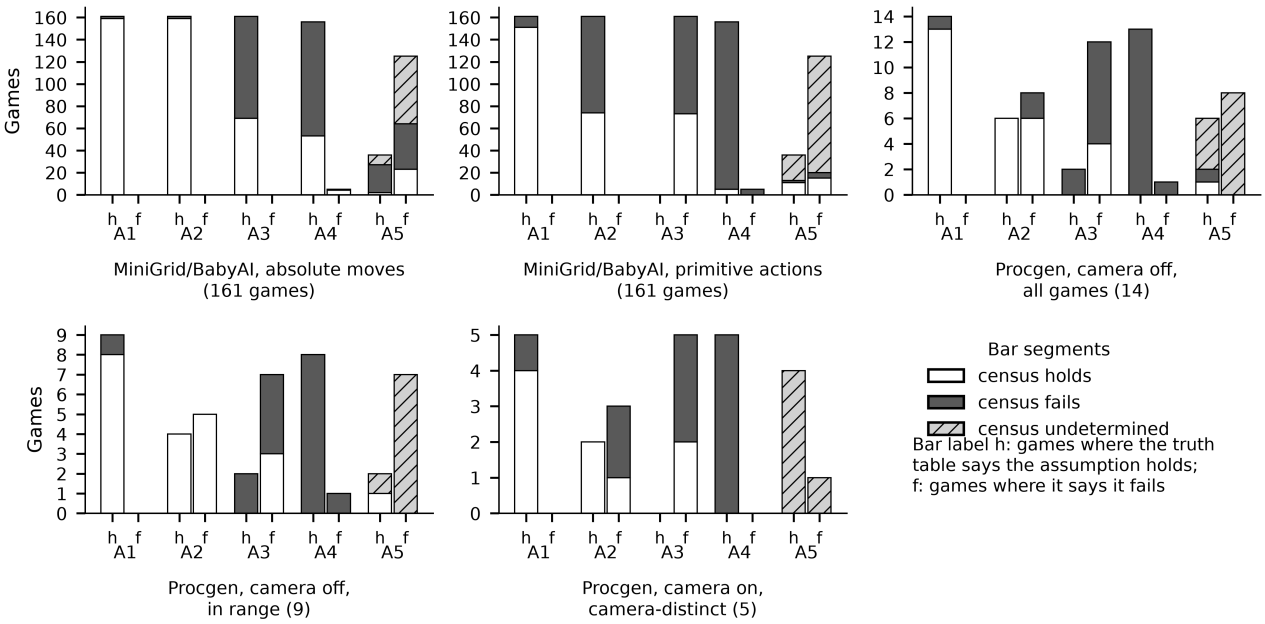


Fig. 3. An exploratory view of V1: for each benchmark and condition and each assumption, the games where the truth table says the assumption holds (bar h) or fails (bar f), split by the census verdict (holds, fails, undetermined). A missing bar is a truth class with zero games. "Camera off" is the whole-level view and "camera on" the view centered on the agent. Exploratory, read after the outcome was known; source in Supplement, Section S.5.

Procgen the counter of walked-through "walls" is above zero on some seeds of all 13 games where the truth holds A4.

For A5, the exploratory counts show that most seeds with an undetermined verdict never recorded a goal place at all: 620 of 663 under absolute moves, 986 of 1,074 under primitive actions and 70 of 110 on Procgen. A place is recorded only at a level end, so on these seeds the census had nothing to test. The census said "holds" where the truth fails on 23 games under absolute moves and 15 under primitive actions; success there needs facing the target, a toggle or a pickup, the level ends on a cell beside or in front of the target, and the census, reading places only, recorded that cell as a goal. The negative kappa values for A5 come mostly from "undetermined", which the registered kappa maps to the value opposite the truth. The cell "truth fails, census holds" used by kappa therefore contains 23 census "holds" and 61 undetermined games under absolute moves (84 in all), 15 and 105 under primitive actions (120 in all), and 0 and 8 on Procgen (8 in all).

Seeds often disagree with their game's majority where the verdict depends on the course of the planner's run (Fig. 4): at least one seed dissents on 134 of 161 games for A2 under primitive actions, 135 for A3 under primitive actions, and 111 each for A3 and A4 under absolute moves, but only on 3 and 6 games for A1 and A2 under absolute moves.

Taken together, these exploratory counts point to a single problem. For A2 under primitive actions, and for A3, A4 and A5, the census probe and the truth definition are not the same construct. The census A2 asks whether some key moves one located object; the truth asks whether exactly one object's displacement is a fixed function of the action, turns included. The census A3 asks whether the planner's first-level model mispredicted; the truth asks whether each action's unblocked displacement is constant. The census A4 counts walked-through "wall" colors; the truth asks whether every blocked move is explained by a fixed obstacle set. The census A5 is silent when no level ends; the truth asks whether success is triggered by entering a place. In each case the census reads a by-product of one planner's run, while the truth reads the environment's step function. This describes the mismatch after the outcome was known; it proposes no fix and adds no result. On the 17 held-out ARC-AGI-3 games, the exploratory V2 line computed by the same frozen functions gives AUROC 0.3542 for N and 0.3403 for R (Supplement, Section S.7); it is reported and never tested. None of this exploratory material supports a causal claim; these are tallies of the planner's internal counters, made after the outcome was known.

VI. Discussion

The registered answer is no on both counts: the census failed the registered validity test on games from the two other benchmarks, and its score passed the registered prediction test for no agent there.

A. What the Failure Means

The truth table is the registered answer key, committed before registration with the source line behind each entry; the census verdict is the instrument under test. A failing V1 line therefore says that the census as registered does not measure that assumption on that benchmark. A V1 failure on any benchmark decides the outcome by itself, and a prediction built on a score that does not measure what it names would be hard to interpret even if it had passed.

The census kept every parameter fixed across benchmarks, which was the registered meaning of transfer, and it still failed.

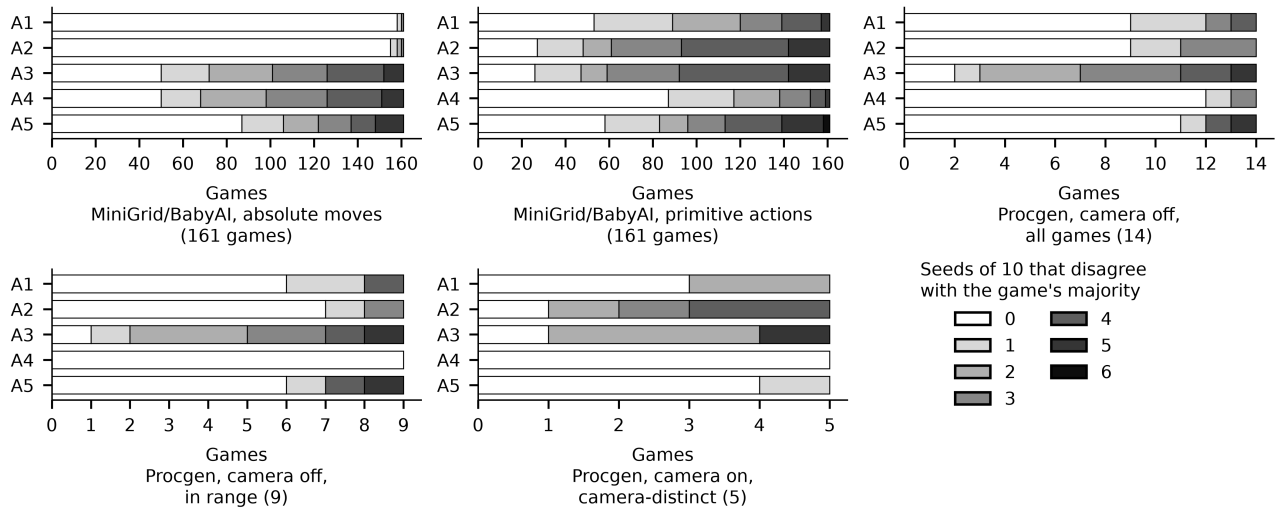


Fig. 4. An exploratory view of seed dissent: for each benchmark and condition and each assumption, the games split by how many of their 10 seeds give a census verdict different from the game's majority verdict. "Camera off" is the whole-level view and "camera on" the view centered on the agent. Exploratory, read after the outcome was known; source in Supplement, Section S.5.

Fixing the parameters keeps the procedure the same; it does not make the procedure measure the same property on new games. If intelligence is defined over a range of environments [8], a claim that an instrument describes games in general is a claim about a range too, to be tested on each part of the range it names; and an instrument that characterizes games carries priors of its own [9], here the planner whose by-products the census read. The registered decision rule worked as designed, and no reading of the results after the fact could turn the refutation into support.

B. Why the Failure Could Not Show on the Development Benchmark

I set the census constants while developing the planner on eight ARC-AGI-3 games, and I wrote the per-seed rules on top of the planner's report. I never wrote the registered development truth table for those games, so V1 was never computed there. The census was tuned to make the planner work, and nothing on ARC-AGI-3 measured whether its verdicts matched the games' mechanics, so a disagreement could not show there. The registered phrase that ARC-AGI-3 "does not count toward generality" kept the development games out of the evidence for the claim, but nothing in it prevented checking the census on them.

C. The Pilot Check That Was Not Done

The pilot runs contained census verdicts for eight games with truth-table entries, six MiniGrid/BabyAI environments and two Procgen games. Comparing them with the truth before the v1.1 tag could have shown the disagreement before the 10,500 confirmatory runs were made; the protocol did not register such a check, and I did not make one. Since the pilot could change adapter code but no census constant, threshold or truth entry, an early warning could not have rescued the claim, but it would have changed what was run next.

D. The Nuisance Baseline

The baseline is fitted in-sample and the census score is not fitted to the outcome at all, so the comparison favors the baseline; it is the comparison the protocol registered. For the agents and conditions examined, how large and how varied a MiniGrid/BabyAI environment is predicted first-level completion better than the five assumptions as measured. That is a statement about the measured census, not about the assumptions as properties of the games.

E. The Methodological Mistake

The reason the census did not carry over is a mistake of method, read from the exploratory diagnosis: the probes and the truth predicates were not the same constructs, and no test existed where the parameters were set. The census decisions were by-products of one planner's run; the truth entries are properties of the environment's step function. Only A1 is measured in nearly the same terms on both sides, and only A1 had no failing V1 line. I wrote the truth definitions and the census rules separately and never checked on any game, before registration, that one estimated the other.

F. What a Second Registered Study Would Have to Change

These are requirements for a different study, not a result and not an existing instrument. It would have to define each probe and its truth predicate as the same construct, write a truth table for the development instance and compute V1 there, all before registration; register a V1 gate with a stop rule on the pilot games of every benchmark; register how "undetermined" enters each test, so that a silent instrument and a wrong one are told apart; and name in advance a nuisance baseline that the census must beat on the development instance.

G. What This Study Does Not Show

This study says nothing about the ability of frame-only agents on these games, and it does not show that A1 to A5 are poor descriptions of the games; the truth tables say how often each holds. It says nothing about the hidden ARC-AGI-3 games or about agents that bring priors to the games. It does not show that a differently built census would fail, or that instruments built on one benchmark cannot transfer to others. It shows only that this instrument failed the tests registered for it.

VII. Conclusion

The registered objective was a frame-only census, built and frozen on ARC-AGI-3, that would report the mechanics of games from other benchmarks at a registered accuracy and predict where assumption-driven agents complete the first level. It failed in each registered respect: V1 failed on both confirmatory benchmarks, for three or four of the five assumptions depending on the benchmark and condition; V2 passed for no agent on any benchmark; and on MiniGrid/BabyAI a baseline built from the size and variety of the environments ranked them better than the census did. The registered consequence followed: a refutation. Freezing the parameters did not make the census valid on the new benchmarks, and I never ran the check that would have shown this before the confirmatory runs: a comparison with known truth on the development and pilot games.

Data and Code Availability

All data and code are public under the MIT license. The data are the 10,500 confirmatory rows, one JSON file per benchmark, condition, agent, game and seed, the 500-row ARC-AGI-3 baseline, the pilot rows, the outputs of the four checks, the analysis logs and the V1 and V2 tables. The code is every script that produced them and the scripts that build and check this article's numbers and figures; the agent I1 keeps the upstream MIT notice of its authors. The registered protocol is fixed at three git tags, prereg-v1, prereg-v1.1 and prereg-v1.2, and in two public registrations on the OSF. An earlier version of this article and the supplement are on Zenodo under CC-BY 4.0. The repository address, the commit hashes, the registration addresses and the Zenodo identifiers are withheld for double-anonymous review and will be supplied to the Editor-in-Chief on request.

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References

[1] D. Perez-Liebana *et al.*, "The 2014 General Video Game Playing Competition," *IEEE Trans. Comput. Intell. AI Games*, vol. 8, no. 3, pp. 229–243, Sep. 2016, doi: 10.1109/TCIAIG.2015.2402393.

[2] ARC Prize Foundation, "ARC-AGI-3: A new challenge for frontier agentic intelligence," Tech. Rep., Apr. 22, 2026. Accessed: Oct. 24, 2026. [Online]. Available: https://arcprize.org/media/ARC_AGI_3_Technical_Report.pdf

[3] M. Chevalier-Boisvert *et al.*, "Minigrid & Miniworld: Modular & customizable reinforcement learning environments for goal-oriented tasks," in *Advances in Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 73383–73394, doi: 10.52202/075280-3209.

[4] M. Chevalier-Boisvert *et al.*, "BabyAI: A platform to study the sample efficiency of grounded language learning," in *Proc. Int. Conf. Learn. Representations*, 2019. [Online]. Available: https://openreview.net/forum?id=rJeXCo0cYX

[5] K. Cobbe, C. Hesse, J. Hilton, and J. Schulman, "Leveraging procedural generation to benchmark reinforcement learning," in *Proc. 37th Int. Conf. Mach. Learn.*, H. Daumé III and A. Singh, Eds., Proc. Mach. Learn. Res., vol. 119, 2020, pp. 2048–2056. [Online]. Available: https://proceedings.mlr.press/v119/cobbe20a.html

[6] M. Genesereth, N. Love, and B. Pell, "General game playing: Overview of the AAAI competition," *AI Mag.*, vol. 26, no. 2, pp. 62–72, 2005, doi: 10.1609/aimag.v26i2.1813.

[7] D. Perez-Liebana, J. Liu, A. Khalifa, R. D. Gaina, J. Togelius, and S. M. Lucas, "General Video Game AI: A multitrack framework for evaluating agents, games, and content generation algorithms," *IEEE Trans. Games*, vol. 11, no. 3, pp. 195–214, Sep. 2019, doi: 10.1109/TG.2019.2901021.

[8] S. Legg and M. Hutter, "Universal intelligence: A definition of machine intelligence," *Minds and Machines*, vol. 17, no. 4, pp. 391–444, 2007, doi: 10.1007/s11023-007-9079-x.

[9] F. Chollet, "On the measure of intelligence," 2019, arXiv:1911.01547.

[10] P. Bontrager, A. Khalifa, A. Mendes, and J. Togelius, "Matching games and algorithms for general video game playing," in *Proc. AAAI Conf. Artif. Intell. Interactive Digit. Entertainment*, vol. 12, no. 1, 2016, pp. 122–128, doi: 10.1609/aiide.v12i1.12884.

[11] R. D. Gaina, S. M. Lucas, and D. Perez-Liebana, "General win prediction from agent experience," in *Proc. 2018 IEEE Conf. Comput. Intell. Games (CIG)*, 2018, pp. 1–8, doi: 10.1109/CIG.2018.8490439.

[12] C. Laidlaw, S. J. Russell, and A. Dragan, "Bridging RL theory and practice with the effective horizon," in *Advances in Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 58953–59007, doi: 10.52202/075280-2573.

[13] E. Rudakov, J. Shock, and B. U. Cowley, "Graph-based exploration for ARC-AGI-3 interactive reasoning tasks," 2025, arXiv:2512.24156.

[14] K. H. Liew, "Explore before you solve: The speed–depth trade-off in epistemic agents for ARC-AGI-3," 2026, arXiv:2605.25931.

[15] A. Skoutnev, K. Acharya, G. Longhitano, M. Udell, K. Ellis, and I. Drori, "Twin: Playing an unknown game with a test-time digital twin," 2026, arXiv:2608.14490.

[16] M. G. Bellemare, J. Veness, and M. Bowling, "Investigating contingency awareness using Atari 2600 games," in *Proc. AAAI Conf. Artif. Intell.*, vol. 26, no. 1, 2012, pp. 864–871, doi: 10.1609/aaai.v26i1.8321.

[17] J. Choi *et al.*, "Contingency-aware exploration in reinforcement learning," in *Proc. Int. Conf. Learn. Representations*, 2019. [Online]. Available: https://openreview.net/forum?id=HyxGB2AcY7

[18] A. Blum and M. Hardt, "The Ladder: A reliable leaderboard for machine learning competitions," in *Proc. 32nd Int. Conf. Mach. Learn.*, F. Bach and D. Blei, Eds., Proc. Mach. Learn. Res., vol. 37, 2015, pp. 1006–1014. [Online]. Available: https://proceedings.mlr.press/v37/blum15.html

[19] R. Roelofs *et al.*, "A meta-analysis of overfitting in machine learning," in *Advances in Neural Inf. Process. Syst.*, vol. 32, 2019. [Online]. Available: https://papers.nips.cc/paper_files/paper/2019/hash/ee39e503b6bedf0c98c388b7e8589aca-Abstract.html

[20] L. Bertinetto, J. F. Henriques, S. Albanie, M. Paganini, and G. Varol, "Preface," in *NeurIPS 2020 Workshop on Pre-registration in Machine Learning*, L. Bertinetto, J. F. Henriques, S. Albanie, M. Paganini, and G. Varol, Eds., Proc. Mach. Learn. Res., vol. 148, 2021, p. i. [Online]. Available: https://proceedings.mlr.press/v148/bertinetto21a.html

[21] S. A. Goldman and M. J. Kearns, "On the complexity of teaching," *J. Comput. Syst. Sci.*, vol. 50, no. 1, pp. 20–31, 1995, doi: 10.1006/jcss.1995.1003.

[22] T. M. Mitchell, "Generalization as search," *Artif. Intell.*, vol. 18, no. 2, pp. 203–226, 1982, doi: 10.1016/0004-3702(82)90040-6.

[23] S. Keren, A. Gal, and E. Karpas, "Goal recognition design," in *Proc. Int. Conf. Automated Planning and Scheduling*, vol. 24, no. 1, 2014, pp. 154–162, doi: 10.1609/icaps.v24i1.13617.

[24] C. Wayllace, P. Hou, W. Yeoh, and T. C. Son, "Goal recognition design with stochastic agent action outcomes," in *Proc. 25th Int. Joint Conf. Artif. Intell. (IJCAI-16)*, 2016, pp. 3279–3285. [Online]. Available: https://www.ijcai.org/Proceedings/16/Papers/464.pdf